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Differentiating Visual Usability from Aesthetics in Visual Experience

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ABSTRACT

Numerous studies have examined the dynamics of visual experience in HCI. Yet findings, particularly regarding the aesthetic–usability effect, remain inconsistent. Moreover, the cognitive and affective processes underlying visual experience have not been explicitly examined or operationalized in existing measurement scales. Reliable measures of these dimensions are needed to understand visual experience formation. To address this gap, we develop and validate a novel scale of *visual usability* as a distinct cognitive dimension of visual technology experience. Scale construction is based on a literature review and exploratory factor analysis of survey data ($N = 660$), with construct validity further assessed through an experiment manipulating visual complexity ($N = 24$). The results indicate that the scale effectively differentiates visual usability as a cognitive, sensemaking dimension of visual experience, through perceptions of clarity, unity, and color. The scale can be used to examine visual technology experience and the aesthetic–usability effect.

KEYWORDS

Visual usability; perceived usability; aesthetics; visual experience; scale

1. Introduction

Vision plays a central role in human-computer interaction (HCI). Through sight, we make sense and experience technological artifacts in numerous ways. The formation of experience through sight in HCI is a complex phenomenon, encompassing cognitive processes such as attention, spatial reasoning (e.g., Palmer, 1999), perceived usability, and affective processes such as appraisal, valence (Russell, 2003), and aesthetic appeal. Visual experiences are influenced by perceptual features of the stimuli such as symmetry, color, and contrast, illustrating how cognition and affect jointly shape the formation of visual experience (Leder et al., 2004).

The complexity of visual experience formation has prompted extensive research into visual technology–related experiences, their underlying dimensions, and their interactions. Consequently, numerous concepts are used to conceptualize visual experience in HCI (e.g., perceived usability, apparent or expected usability, perceived aesthetics, aesthetic appeal, and beauty), and corresponding measurements have been developed. However, the conceptual overlaps of these concepts remain unclear, and the cognitive sensemaking dimension of visual experience has remained largely underexplored. Therefore, we develop and validate a distinct scale of visual usability that addresses the cognitive dimension of visual experience.

Although extensive research has examined user experience (UX) broadly, understanding of its distinct dimensions and their interrelations remains limited (Hornbæk & Hertzum, 2017). Likewise, the underlying dimensions of visual experience, a subtype of UX focused on experience formation through sight, require further conceptual clarification and reliable measurement. In HCI research, aesthetic experience is often treated as a single construct, despite visual and aesthetic experiences inherently involving both cognitive and affective processes (e.g., Carroll, 2001; Di Dio & Gallese, 2009; Silvennoinen & Jokinen, 2016a).

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Figure 1. The hierarchy of the visual experience concepts in HCI examined in the current study.

The key to better understanding visual experience is to examine its cognitive and affective processes in detail (Di Dio & Gallese, 2009). While these processes are intricately interwoven, they can be analytically distinguished to enable more precise investigation. Identifying and measuring the underlying dimensions of visual experience can improve the predictability of experience design and help explicate conflicting findings of the aesthetic-usability effect which often arise from divergent conceptualizations and operationalizations of perceived usability and aesthetics (e.g., Hassenzahl & Monk, 2010; Lima & Gresse von Wangenheim, 2022; Silvennoinen & Jokinen, 2016a). Accordingly, the first goal of this study is to differentiate visual usability from aesthetic appeal within visual experience.

Visual usability operates as a cognitive sensemaking vehicle in visual technology experiences (Gretalita et al., 2017; Shao & Wang, 2018; Silvennoinen et al., 2014, 2017; Silvennoinen & Jokinen, 2016a) by focusing on how visual elements (e.g., lines, colors, visual rhythm, and symmetry) in visual user interfaces (UIs) are mentally represented and how visual elements relate to the functions afforded by them (Silvennoinen & Jokinen, 2016a, 2016b). Thus, the focus of visual usability is on how fluently and intuitively visual information can be processed, interpreted, and understood in terms of represented goals and how visual elements guide the interaction (Schlatter & Levinson, 2013; Silvennoinen et al., 2014). Several scales for measuring visual experience in HCI ranging from perceived usability to aesthetic appeal have been developed but there is a lack of a distinct and a reliable scale for measuring cognitive dimension in visual experience of technological artifacts. Therefore, the second goal is to operationalize and validate a novel scale of visual usability to capture the more cognitively- and task-oriented way of seeing underlying visual experience formation. The concepts utilized in HCI to examine visual experience, and the scope of this paper are presented in Figure 1.

In measuring visual experience, it is essential to distinguish the more objective sensemaking aspects of visual usability from the more individually variable aesthetic appeal. Differentiating visual usability from perceived usability and aesthetic appeal, and validating a visual usability scale, enables future research to examine the aesthetic-usability effect in greater detail, clarify the underlying dimensions of visual experience, and investigate their interactions in HCI. This supports a more precise understanding of the impact of visual usability on visual technology experience and better informs visual design decisions. In addition, this differentiation increases predictability in experience design, as visual usability is assessed more consistently than aesthetic appeal and can be used to increase the likelihood of meaningful aesthetic appeal (Silvennoinen & Jokinen, 2016a).

In summary, the key contributions of this study to visual experience formation in HCI are:

- We create and validate a novel scale (visual usability) that captures cognitive dimensions of visual experience in technological artifacts, supports analysis of the aesthetic-usability effect, and informs visual interface design.
- We validate the scale using exploratory factor analysis with 660 respondents and make it available for use by HCI researchers and practitioners (Supplementary Appendix A Table 2).
- We further validate the scale's construct validity through an experiment involving visual stimulus complexity manipulations with 24 participants.
- Our findings indicate the scale's effectiveness in differentiating cognitive dimensions of visual experience from the affectively oriented perceptions of aesthetic appeal.

The paper is structured as follows. First, previous research on visual experience in HCI research is reviewed in terms of methodological and conceptual considerations and existing measurements.

Second, selection and operationalization of the candidate constructs for the visual usability measurement (based on a literature review) are presented along with two pilot studies. Third, findings of a survey study with exploratory factor analysis and analysis of variance are presented together with the final factor solution. The results of the validation study are then discussed, indicating the novel scale's ability to measure visual usability efficiently. Lastly, the conclusions of the study are presented with future research topics.

2. Related work

Visual UIs communicate aesthetic, symbolic, and affective meanings that extend beyond their role as task-completion and goal-oriented devices (e.g., Crilly et al., 2004; Krippendorff, 1989, 2006; Postrel, 2003). Thus, technological artifacts are valued not only for their functional properties but also for their aesthetic and emotional dimensions in UX (e.g., Desmet & Hekkert, 2007; Hassenzahl, 2001; Lavie & Tractinsky, 2004). One of the most examined dimensions of visual experience is the interplay between aesthetics and perceived usability (e.g., Hassenzahl & Monk, 2010; Lee & Koubek, 2010; Minge & Thüring, 2018; Thüring & Mahlke, 2007). Research on the aesthetic-usability effect (aesthetic appeal increasing perceived usability, Lidwell et al., 2010), gained scholarly attention following early work on apparent usability (Kurosu & Kashimura, 1995) and the "What is beautiful is usable" article Tractinsky et al., 2000). This line of research, referred to as visual aesthetics in HCI (e.g., Tractinsky, 2013), examines aesthetics in HCI, the dynamics among perception-based experience concepts, and aims to improve the predictability of visual design decisions targeting specific technology experiences (Jokinen et al., 2018).

2.1. Conceptual and methodological considerations

Previous research has acknowledged conflicting findings on the relationship of perceived usability and aesthetics stemming from conceptual and methodological inconsistencies (e.g., Hassenzahl & Monk, 2010; Lima & Gresse von Wangenheim, 2022; Silvennoinen & Jokinen, 2016a). Studies of this interplay use divergent definitions and methods of perceived usability (Tractinsky, 2020), including perceived usability (e.g., Hassenzahl & Monk, 2010; Sonderegger & Sauer, 2010), apparent usability (e.g., Kurosu & Kashimura, 1995), and expected usability (Thielsch et al., 2015). Perceived usability has also been used to describe subjective usability during actual use (Thüring & Mahlke, 2007), to contrast with objective usability metrics such as task completion time (Sonderegger et al., 2012), as an equivalent of pragmatic quality (Hassenzahl & Monk, 2010), and as a synonym for user satisfaction (Veral & Macías, 2019). Here, we refer to perceived usability as subjective perceptions of usability of an artifact before its actual use (e.g., Tractinsky et al., 2000), in which perceived usability is defined as users' subjective perceptions and judgments of its usability (e.g., Hassenzahl, 2004; Thüring & Mahlke, 2007).

The aesthetic-usability effect has been found to operate in various ways. Aesthetic appeal can increase the understandability of visual UIs (Hassenzahl & Monk, 2010; Lindgaard & Dudek, 2003; Tractinsky et al., 2000), positively influence interaction (Moshagen et al., 2009), enhance performance in demanding tasks (Norman, 2004), increase or decrease performance, and motivate continued interaction with an artifact (Sonderegger & Sauer, 2010). Moreover, evaluations of aesthetic appeal can be formed quickly and play a key role in first impressions of encountered artifacts (Lindgaard et al., 2006; Papachristos & Avouris, 2011). Findings of aesthetic appeal on performance are mixed, with both positive and negative outcomes reported (Hartmann et al., 2007; Moshagen et al., 2009; Sonderegger & Sauer, 2010).

In the quest of explicating the aesthetic-usability effect moderating factors have been proposed, such as interaction medium, goal orientation, task type, performance measures, aesthetics manipulation, and aesthetics measures have small but heterogeneous effects on user performance (Thielsch et al., 2019b). Other examined moderators include overall evaluations of an artifact's "goodness" (Hassenzahl, 2004; Hassenzahl & Monk, 2010; van Schaik & Ling, 2008), usage context, product category (e.g., Sonderegger et al., 2014), and goal orientation in learning versus performance goals (Thielsch et al., 2019a). Halo effect has also been reported to affect the aesthetic-usability effect bidirectionally, hedonic halo effect prior to interaction and pragmatic halo effect during interaction (Minge & Thüring,

2018). More recently, visual clarity has been proposed as a moderator or mediator (Schrepp et al., 2020). However, further research is needed to identify the psychological processes and moderators of aesthetic-usability effect and to develop reliable methods for examining these mechanisms (Schrepp et al., 2020). For example, user identification with an artifact, artifact characteristics, and sensory dominance have been proposed as additional influencing factors (Sauer & Sonderegger, 2022).

Several HCI studies adopt an objectivist approach to visual experience by considering visual elements of technological artifacts as determinants of aesthetic appeal (e.g., Bauerly & Liu, 2006; Kim et al., 2003; Lin et al., 2013; Ngo & Byrne, 2001; Tuch, Presslaber, et al., 2012a; Tuch, Roth et al., 2012b). In contrast, subjectivist approaches emphasize top-down processes of the perceiver, viewing visual experience as constructed by the observer (e.g., Lavie & Tractinsky, 2004; van der Heijden, 2003). However, visual experience emerges in the interactive process between the objectivist and the subjectivist approaches (Jokinen et al., 2018; Reber et al., 2004: Silvennoinen, 2017; Silvennoinen, 2021). Visual experience is a mental phenomenon including mental information contents and involves multiple underlying cognitive and affective processes affected by visual entities constructed of visual elements (e.g., Silvennoinen & Jokinen, 2016a).

Operationalizations of aesthetic appeal (synonyms e.g., perceived visual and beauty) also vary considerably in HCI research and is often treated as one-dimensional construct, with scales such as unattractive–attractive (e.g., Lindgaard et al., 2006), low–high aesthetics (e.g., Alharoon & Gillan, 2020; Tractinsky et al., 2000), non-appealing–appealing (e.g., Sonderegger & Sauer, 2010), or pleasant–unpleasant (e.g., Thielsch & Hirschfeld, 2012). This reductionist approach reflects a lack of foundational analysis of visual experience in HCI and the coexistence of parallel pluralist perspectives that do not connect to each other, resulting in fragmented research (Udsen & Jørgensen, 2005). Consequently, key underlying dimensions of visual experience are often overlooked, raising concerns about the reliability of visual experience research in HCI (Schrepp et al., 2020; Silvennoinen, 2021). In contrast, contemporary aesthetic theory conceptualizes visual experience as an interpretative process involving multiple cognitive and affective dimensions (e.g., Carroll, 2001), whereas HCI research on aesthetic appeal often adopts a contrasting view emphasizing immediate responses to visual stimuli (e.g., Marković, 2012; Moshagen & Thielsch, 2010; Santayana, 1955).

2.2. Measurements

Existing measures of technology-related experience are diverse, but none focus specifically on visual usability as defined here. UX measurements typically adopt a holistic perspective on experience formation, with commonly used measurements including AttrakDiff2 (Hassenzahl, 2004; Hassenzahl et al., 2003) and the Usability Metric for User Experience (UMUX; Finstad, 2010). AttrakDiff2 comprises four components: hedonic quality identification, hedonic quality stimulation, pragmatic quality, and evaluative constructs (see Supplementary Appendix A.1 for semantic differentials). Other measures target specific UI aspects; for example, Web-CLIC (Thielsch & Hirschfeld, 2019) assesses content experience through clarity, likeability, informativeness, and credibility.

In research on visual aesthetics, Classical and Expressive Aesthetics (CEA: Lavie & Tractinsky, 2004) and the Visual Aesthetics of Websites Inventory (VisAWI; Moshagen & Thielsch, 2010) are commonly used measures. CEA include two components, classical and expressive aesthetics and were validated against usability, pleasurable interaction, and service quality (Lavie & Tractinsky, 2004). VisAWI comprises four components: simplicity, diversity, colorfulness, and craftsmanship (see Supplementary Appendix A.1 for item lists). Despite their widespread use, these measures do not differentiate visual usability from aesthetics in visual experience but instead treat different modes of visual perception as aesthetic qualities. Aesthetic experience differs from the more neutral concept of visual experience, which does not inherently imply affective qualities. Visual experience is the umbrella term for aesthetic experience (occurring if certain gratifications on sensory, cognitive, and affective conditions are met). Perceived usability captures subjective usability judgments on what is perceived, including pre-use impressions, whereas visual usability focuses on the understandability of visual UI design. Differentiating and explicating these concepts brings us closer to the explication of the underlying dynamics of visual experience in HCI.

Two widely used standardized measures of perceived (general) usability are the Computer System Usability Questionnaire (CSUQ; Lewis, 1995, 2018) and the System Usability Scale (SUS; Brooke, 1996). These measures assess overall post-use perceptions of usability and therefore differ from the scope of visual usability. In addition to questionnaire-based measures, automated tools have been developed to assess UI aesthetics (Miniukovich & De Angeli, 2015), including the visual complexity of web pages (Michailidou et al., 2021). The development of such tools is based on user scores which are compared to computed scores of automatic metrics. Automated tools can enhance research on underlying dynamics of visual experience in HCI. However, these often present results in abstract formats (e.g., visual complexity scores; Michailidou et al., 2021) that require visual design expertise to translate into actionable redesign decisions. In contrast, a visual usability scale could be utilized in identifying visual design aspects that increase or decrease visual usability, thereby directly informing design decisions.

3. Selection and operationalization of the candidate constructs and items for the new scale

The construction of the new scale started with a stand-alone literature review (Okoli & Schabram, 2010) focusing on definitions and existing scales of perceived usability and aesthetic appeal to explicate the current conceptualizations and operationalizations of the phenomenon. Aesthetic as a keyword was also included in the literature review to enable the detection of research articles examining the aesthetic-usability effect and to analyze how these two dimensions have been operationalized. The goal of this procedure was to find the utilized operationalizations of perceived usability and aesthetic appeal and differentiate constructs of visual usability (i.e., cognizing visual UI design in the sensemaking process), and then exclude constructs used to measure aesthetic appeal (i.e., experiential constructs appraising affective qualities). Thus, the selection of the constructs for visual usability was conducted according to its definition: visual elements (e.g., lines, colors, visual rhythm, and symmetry) as sensemaking vehicles in visual UIs, which are mentally represented and indicate functions afforded by them (Silvennoinen & Jokinen, 2016a, 2016b).

Five databases (ACM Digital library, PubMed, EBSCO, IEEE Xplore Digital library and Google Scholar) were utilized as sources with the following keyword combinations: aesthetic + usability + measure; aesthetic + usability + evaluation; visual + usability + measure; visual + usability + evaluation; apparent + usability + measure, and apparent + usability + evaluation. The keywords were to be found in the title, abstract or keywords, to be included in the search results. In addition, the search was further enhanced by citation-based and author-based backward and forward searches.

The qualitative content analysis conducted followed the summative content analysis procedure by first detecting and defining the most central contents and words emerging from the textual data (Hsieh & Shannon, 2005). The focus of this analysis phase was to identify different aspects of visual usability and visual design factors (e.g., related to composition, balance and colors). The articles under analysis were analyzed accordingly. The information on what aspects of visual usability were studied and with what concepts were tabulated. After this, the information was grouped into sections focusing on similar contents.

Overall, the literature searches resulted in 28,439 articles, from which articles fulfilling exclusion criteria were left out of the analysis. The exclusion criteria were applied if the article did not examine the measurement of visual usability and if the measurement utilized was not described in the method section of the article. In addition, articles focusing on measuring information visualization techniques and applications were excluded due to the scope of the review. Moreover, if the article focused on a user group with accessibility issues (e.g., deficits in eyesight), the articles were excluded. In addition to the general search terms, the following inclusion criteria needed to be met for an article to be included in the review: the article deals with measuring visual usability; this is expressed in the title, keywords, or abstract of the article. The literature review resulted in 57 articles. The results indicated that previous research articles approached the phenomenon in general through differing conceptualizations of aesthetic appeal. The most frequent definitions utilized were aesthetics (Ben-Bassat et al., 2006; Cyr et al., 2006; De Angeli et al., 2006; Moshagen et al., 2009; Purchase et al., 2011), beauty (Dupuy-Chessa et al., 2016; Kurosu & Kashimura, 1995; Schenkman & Jönsson, 2000; Tractinsky, 1997), visual attractiveness (Lindgaard et al., 2011; Lindgaard & Dudek, 2003; Reinecke & Bernstein, 2011; Sonderegger & Sauer,

2010), and visual appeal (Stojmenovic et al., 2014). Articles ($n = 16$) that solely focused on aesthetics, emotions (e.g., aesthetic joy) and interactive usability as separate research subjects were excluded from the literature review as the focus of these articles was not on the interplay between aesthetic appeal and perceived usability.

The resulting 41 articles examining perceived usability and aesthetic appeal were analyzed regarding the operationalizations of perceived usability, focusing especially on constructs relating to visual usability and grouped into coherent subcategories reflecting similar contents. The analysis resulted in the identification of 29 distinct visual usability elements and principles, including: color, clarity, cleanliness, symmetry and balance, simplicity and minimalism, readability, layout, structure, placement, order, grouping, contrast, unity and homogeneity, professionalism, diversity and variation, dominance, rhythm, prototypicality, sophistication and harmony, density, fonts and text, images and icons, focal point, size, motion and rhythm, lines and patterns, shapes, continuity, reliability, proximity, and space. When developing the scale items, the guiding principle was to proceed from broader visual usability principles and elements to more detailed components (Silvennoinen & Jokinen, 2016b). For example, clarity, cleanliness, simplicity, and minimalism were grouped into one construct, labeled "visual clarity," which became the first subscale and included seven items. The second subscale focused on layout and was the most extensive, comprising 16 items. The third subscale measured contrast (three items), and the fourth focused on color (four items). The most frequently utilized constructs were colors, contrast, symmetry, balance, layout, clarity, text, patterns, and shapes. The categorization was conducted according to the frequency of the found constructs.

For instance, color was a frequently examined construct (Bhatia et al., 2011; Gronier, 2016; Moshagen & Thielsch, 2010; Reinecke et al., 2013; Uribe et al., 2017) as well as complexity (Deng & Poole, 2010; Reinecke et al., 2013; Reppa & McDougall, 2015), and clarity (Lavie & Tractinsky, 2004; Lin, 2013; Mbipom & Harper, 2011). Of the 41 articles, 12 focused explicitly on constructs pertaining to visual usability. The most used keywords in the 12 articles were visual user interface design, visual user interface, visual design, visual clarity, visual cohesion and visual complexity. The scale items (Table 1, in the next section) were compiled according to these 12 articles. The compilation resulted in the operationalization of visual usability into three components (1) clarity, (2) unity, (3) color and contrast. In the first version of the novel measurement, these components were targeted with 30 items derived from the reviewed literature.

4. Pilot studies and the survey items

Two pilot studies were conducted to analyze the reliability of the targeted components (unweighted mean of scale items) and reduce the number of items and ambiguity, where possible. The first pilot study was conducted as a Webropol web survey (https://www.webropolsurveys.com/) with $N = 30$. The stimulus was a screenshot of a course site in an online learning environment (Figure 2). The survey included 30 items based on the literature review (translated into Finnish). Requests to reply to the survey were sent by email to university mailing lists. The reliability analysis was done with Cronbach's alpha using IBM SPSS Statistics (version 28).

As a result of the first pilot study, it was found that alphas were significantly improved (>0.70) and the survey shortened by removing 14 items that had the greatest decreasing effects on the alphas. Further, based on alphas, it was concluded that color and contrast would be more appropriate to be combined under the same factor than as separate factors. This combination was also influenced by the fact that there were only three items about contrast. With these modifications, the next survey was targeted to measure the following components of visual usability. Clarity was measured by five items, of which alpha was 0.746. The second component was Unity, with an alpha of 0.795, with five items. The third component was Color and contrast and contained six items, with an alpha of 0.723. These components, associated items, and reference sources are presented in Table 1. Some of the items were presented as negative statements to make the respondents pay more attention to their replies and to identify respondents who might provide inconsistent answers.

In the second pilot study, we wanted to ensure further the reliability of the targeted components based on the modifications done after the first pilot study. The stimulus was a screenshot of a web

Table 1. Components, items and reference sources of the original candidate visual usability scale based on literature and two pilot studies.

Component	Items	Reference sources
Clarity	The layout of this website is clear and simple.	(Bhatia et al., 2011; Gronier, 2016, Lavie & Tractinsky, 2004, Lewis, 2002; Seo et al., 2015)
	The website is clearly designed.	(Elling et al., 2007; Lavie & Tractinsky, 2004, Seo et al., 2015; Lewis, 2002)
	The website layout is too dense. (reversed for analysis)	(Bailey & Seals, 2017; Moshagen & Thielsch, 2010)
	The layout of the website is easy to perceive.	(Bhatia et al., 2011; Elling et al., 2007; Lewis, 2002; Miniukovich & De Angeli, 2014; Moshagen & Thielsch, 2010)
	The website has one clear center point where the glance is directed.	(Elling et al., 2007; Lin, 2013; Miller et al., 2012)
Unity	The website is designed to be unified.	(Moshagen & Thielsch, 2010)
	The website is symmetrically designed.	(Lavie & Tractinsky, 2004; Lin, 2013)
	All the elements of the website are interrelated.	(Moshagen & Thielsch, 2010; Lin, 2013)
	The contents of the website are arranged smoothly.	(Lin, 2013)
	The website uses similar forms and lines repeatedly.	(Lin, 2013)
Colors & contrast	The darker part of the website stands out against the brighter part (for example, the dark text stands out from the light background).	(Lin, 2013)
	The colors of the website do not work together. (reversed for analysis)	(Moshagen & Thielsch, 2010)
	The color selection of the website is unsuccessful. (reversed for analysis)	(Moshagen & Thielsch, 2010)
	The website has too many colors. (reversed for analysis)	(Bhatia et al., 2011)
	Colors on the website are pleasing to perceive.	(Moshagen & Thielsch, 2010)
	The different colors of the website make it easier to perceive the content.	(Moshagen & Thielsch, 2010; Lin, 2013)

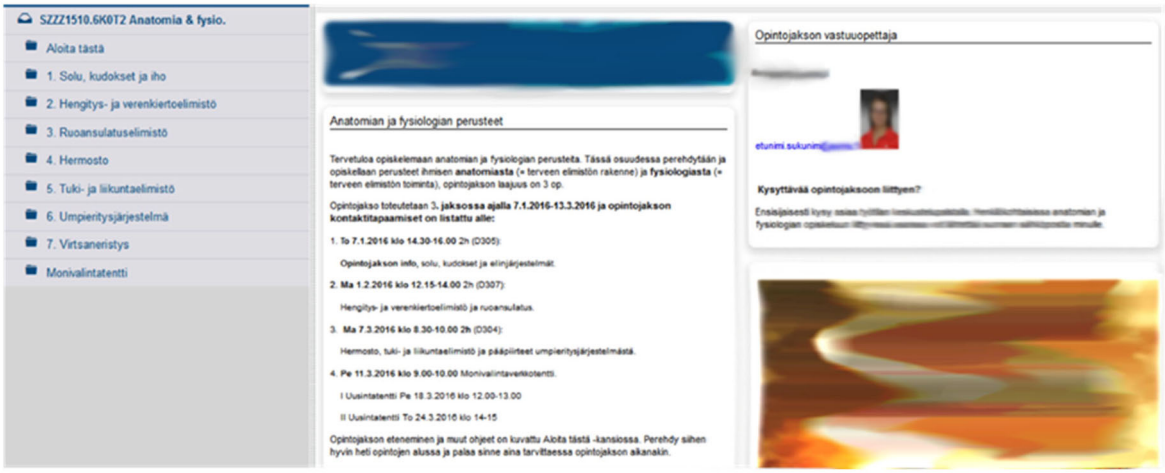


Figure 2. Stimuli for the first pilot study (personal information is blurred for this article due to privacy reasons).

page on the CSS Zen Garden website (Figure 3: Kyoto Forest, John Politowski, http://www.csszengarden.com/207/). The online Webropol survey distributed via email was answered by 31 participants. The study gave further support for the reliability of the components, with alphas 0.889 for Clarity, 0.859 for Unity, and 0.892 for Colors and contrast.

5. EFA and ANOVA

We further developed and analyzed the validity of the novel scale through a survey study with 660 respondents, exploratory factor analysis (EFA) and analysis of variance (ANOVA). Exploratory factor analysis (EFA) was conducted to validate the psychometric structure of the three candidate factors of



Figure 3. Stimuli for the second pilot study http://www.csszengarden.com/207/.

Visual Usability (VU) together with ANOVA by comparing participants' VU and aesthetic experience scores on 33 websites.

First, EFA was conducted to validate the targeted psychometric structure of the developed VU factors and their covariation structure. Further, evaluation data was collected on the same stimuli with three existing scales intended to measure the visual experience related to aesthetic, hedonic and expressive qualities of the stimuli, and this visual experience was then compared to the findings on the average VU score per stimulus. It was expected that the VU scores should deviate significantly from the experience scores especially for simple but esthetically unappealing stimuli. This served to illustrate that the novel VU measurements can differentiate between visual usability and aesthetic experience in website designs.

5.1. Stimuli

The stimuli consist of 33 website landing pages, of which 12 were labeled as esthetically appealing and 11 not esthetically pleasing, respectively. The stimuli can be found in Supplementary Appendix A.3.

The selection of the stimuli started by browsing websites and blogs that rank website designs as beautiful (or best) and unappealing (or worst) with justifications for the rankings. By including both esthetically pleasant and unpleasant websites for the study, we wanted to induce variability in the respondents' visual experiences. The selection criteria also included that the websites are not personal websites of any identifiable individual and that the websites should not be from well-established brands to avoid possible familiarity effects. The sources for the stimuli were selected from the following four websites: www.thomasdigital.com; www.plerdy.com, www.colibriwp.com; and www.ceblog.s3.amazonaws.com.

5.2. Design, apparatus, procedure and participants

The survey scales consisted of the candidate VU items and three previously validated scales from the AttracDiff2 (Marc Hassenzahl et al., 2003; Hassenzahl, 2004), classical and expressive aesthetics (Lavie & Tractinsky, 2004) and VisaWi (Moshagen & Thielsch, 2010). The existing scales were selected based on their intended purpose of measuring the visual experience related to aesthetic, hedonic and expressive qualities of the stimulus. The selected scales were: Hedonic quality – stimulation (HQS), Expressive Aesthetics (EAE) and Diversity (DIV), see Supplementary Appendix A.1.

The participants were recruited via Prolific, an online survey recruitment platform (https://www.prolific.co). Thirty-three different surveys were created with the Webropol survey tool, all with different stimuli. There were no specific restrictions placed on the participants besides that they should be proficient in English and that they should not have replied to any of the 33 surveys before. Thereby there were over 125,500 eligible participants in the Prolific database. Equal gender balance and 20 participants per survey were targeted. Twenty participants per survey resulted in $N = 660$, that is, twenty times the number of VU survey items, which should be a sufficient sample size for EFA. Data from participants who had taken another of our surveys before, based on their Prolific ids, or whose survey completion times were below two and half minutes (i.e., less than 4 s reply time per item, demographical questions included), were rejected.

In total, there were 660 participants with various backgrounds. Their ages varied from 17 to 68 years ($M = 28.3$, $SD = 9.3$) and there were 323 female, 326 male and 11 non-binary participants by gender. Two-hundred and twenty-two (33.6%) of the participants were native English speakers. They were from 32 different countries located in Africa, Asia, Europe, North America and South America. The participants were rewarded by £17.02 average reward per hour. In general, the survey studies did not fulfill our university's criteria for studies to be reviewed by the institutional review board.

5.3. Analysis

The data was preprocessed in MS Excel (version 16.76) and IBM SPSS (version 28), and the demographic variables were analyzed in Excel. The EFA model was defined and analyzed with IBM SPSS (v28). There was no missing data.

5.4. Results

Exploratory Factor Analysis (EFA) was conducted in SPSS version 28 using the Maximum likelihood extraction method with Promax rotation. The oblique rotation method was selected based on the assumption that the underlying factors influencing a visual usability could, and should, correlate (Pituch & Stevens, 2016). Bartlett's test of sphericity indicated significant intercorrelations among the components of the survey items, $\chi^2(120) = 2945.481$, $p < 0.001$. Additionally, the Kaiser-Meyer-Olkin test of sampling adequacy confirmed that the data were suitable for EFA (KMO: 0.925). The average communality value was .537. Factors with Eigenvalues exceeding 1 were retained in the factor solution. Further, parallel analysis was conducted to ensure that the number of factors fitted the data (O'Connor, 2000; Pituch & Stevens, 2016).

In each iteration, items with factor loadings below 0.40 or with cross-loadings above 0.40 in the pattern matrix were excluded, one at a time (Pituch & Stevens, 2016). The only item targeted to measure perception of contrast ("The darker part of the website stands out against the brighter part…") was excluded as it did not load to any of the factors with over 0.40 loading. This item was focused on the contrasts on the web page whereas the other color-related items focused more on color design. Contrast assessment might therefore require its own factor. Further, the item "The contents of the website are arranged smoothly" had a loading of 0.524 to the Clarity factor, instead of the Unity factor, and was therefore included in this factor instead of Unity. This change makes theoretically sense as smoothness of arrangement of visual elements can be seen as an attribute related to unity and not pertaining to evaluation of clarity.

After two iterations, a total of three factors emerged in the final solution based on Eigenvalues >1, with rotation converging in five iterations. To assess the suitability of the solution to the data, parallel analysis (O'Connor, 2000) confirmed that three is a proper number of factors to account for the covariances in the data, as three factors had Eigenvalues higher than the generated 95th percentile Eigenvalues from 1000 generated datasets (Pituch & Stevens, 2016). No cross-loadings exceeding 0.40 were observed in the final pattern matrix. The resulting factor loadings for the included items per factor in the final solution can be found in Table 2. The final scale is also included in the Supplementary Appendix A2.

Table 2. Factors, items and factor loadings per item (pattern matrix, ordered by strength) of the final visual usability scale.

Factor		Items	Factor loading
Clarity	1.	The layout of this website is clear and simple.	1.012
	2.	The layout of the website is easy to perceive.	0.760
	3.	The website layout is too dense.	−0.755
	4.	The website is clearly designed.	0.725
	5.	The contents of the website are arranged smoothly.	0.524
	6.	The website has one clear center point where the glance is directed.	0.410
Unity	7.	All the elements of the website are interrelated.	0.748
	8.	The website uses similar forms and lines repeatedly.	0.697
	9.	The website is symmetrically designed.	0.427
	10.	The website is designed to be unified.	0.400
Colors	11.	The color selection of the website is unsuccessful.	−0.974
	12.	The colors of the website do not work together.	−0.866
	13.	Colors on the website are pleasing to perceive.	0.780
	14.	The different colors of the website make it easier to perceive the content.	0.587
	15.	The website has too many colors.	−0.562

Table 3. Correlations between the VU factors.

	Clarity	Unity	Colors
Clarity	1.00		
Unity	0.568	1.00	
Colors	0.673	0.720	1.00

Factor scores were computed and recorded for the three factors by using regression. Correlations between the factors are shown in Table 3. The strong correlations suggest that the independent factors can be combined together as a sum variable representing Visual Usability (VU) as perceived by the respondent. Cronbach's alpha for the combined VU variable was 0.89.

Based on the final factor model, the evaluations of the stimuli were more closely investigated with two sum variables, other for the visual usability (VU: mean of the standardized factor scores) and the other for the aesthetic experience (AES: mean of the standardized HQS, EAE and DIV scores, Cronbach's alpha = 0.88, Pearson correlations: HQS-EAE: $r = 0.78$, HQS-DIV: $r = 0.62$, EAE-DIV: $r = 0.76$). Pearson's correlation between VU and AES was 0.55. Repeated measures ANOVA with measurement as the within-subject and stimulus as the between-subject factor indicated a significant effect on the evaluation score by stimulus, $F(32,627) = 7.586$, $p < 0.001$, $\eta_p^2 = 0.450$. The interaction between measurement and stimulus was also significant, $F(32,297) = 4.317$, $p < 0.001$, $\eta_p^2 = 0.317$.

All the stimuli are ordered by their VU and AES scores for comparison in Supplementary Appendix A.3. Figure 4 illustrates the differences in the respective VU and AES scores per stimulus. While there are similar patterns in the changes of the VU and AES scores by stimulus, with some stimuli there are notable relative and absolute differences. The most notable discrepancies in the VU and AES scores are for stimuli #7, #9, #23 and #31 (Figures 4 and 5). VU is scored significantly higher than AES for #23 and #31. These findings are understandable as websites' #23 and #31 visual usability (clarity and unity in particular) can be easily cognized, but their designs were a priori categorized as esthetically unappealing. Websites' #7 and #9 AES are scored close to the mean (0) of all websites, and both were categorized as esthetically unpleasant before the study but their VUs are scored significantly lower than zero. The designs seem to be crowded with information, which may explain the VU effect. On the other hand, both VU and AES scores are very similar, above-average levels for stimuli #3, #5, #17, #20 and #22. Again, this makes sense as all the websites are similar in that they seem relatively simple and these were all categorized as esthetically appealing before the study. Stimuli #18, #19 and #21 all had very low VU and AES mean scores. These were all rated unaesthetically unpleasant a priori, and the designs look highly complex, cluttered and dispersed. Clear differences in the rankings of the stimuli by VU and AES are visible also in the Supplementary Appendix A.3. Further, as a note, there is more variation in the AES scores than in the VU scores per stimuli, as indicated by the lengths of the 95% CIs of means in Figure 4.

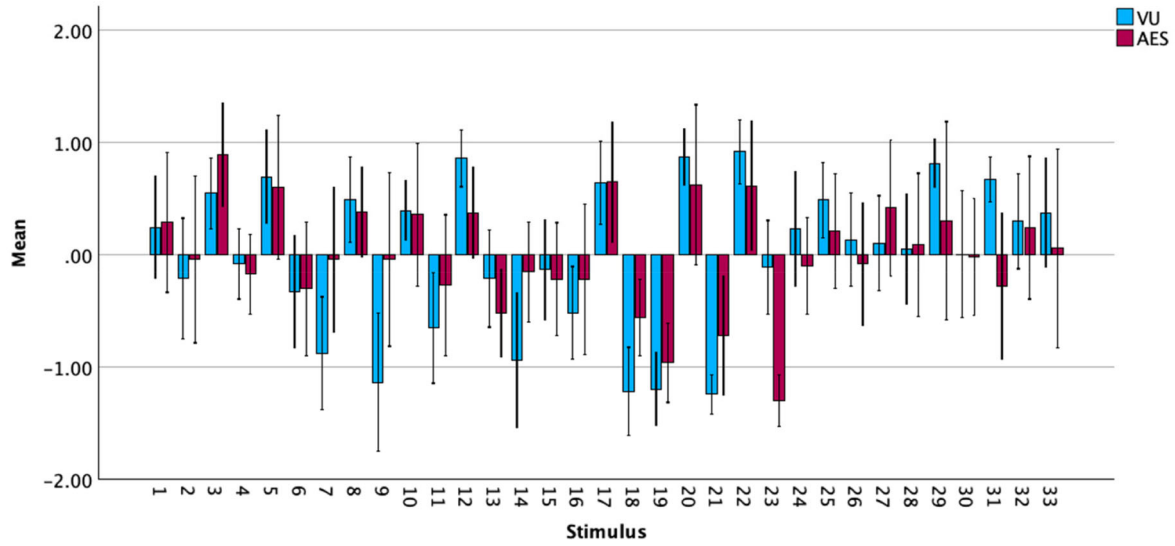


Figure 4. Variations in standardized VU and AES scores by stimulus (website). Error bars: 95% CI.

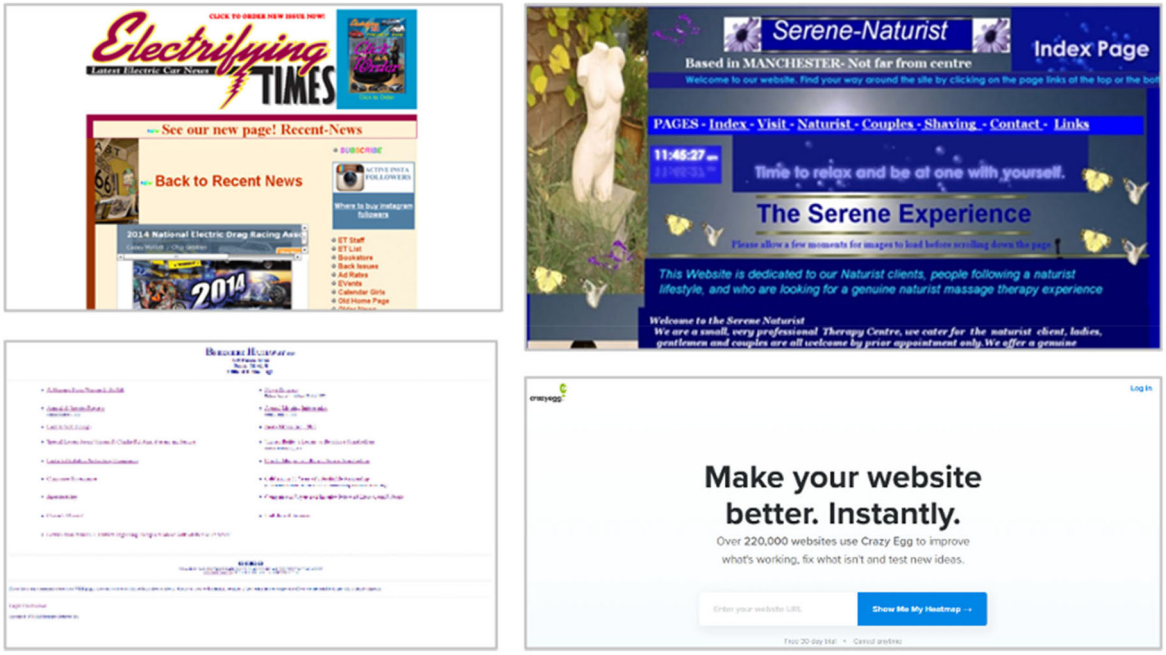


Figure 5. First row: Stimulus #7 and #9: Low VU/average AES (both). Second row: Stimulus #23: Average VU/low AES, and stimulus #31: High VU/low AES.

6. Effects of visual complexity on visual usability

Construct validity of the new visual usability scale was further analyzed by comparing its results with the results on the same two stimuli obtained with the three most well-known and cited scales for measuring visual aesthetics – including items related to visual usability. The scales were VisaWi (Moshagen & Thielsch, 2010), classical and expressive aesthetics (Lavie & Tractinsky, 2004), and AttracDiff2 (Hassenzahl, 2004; Hassenzahl et al., 2003).

6.1. Stimuli

Perceived visual usability has been associated with visual complexity (e.g., Deng & Poole, 2010), as visual complexity is a factor that decreases perceived visual usability. Therefore, the stimuli for the study



Figure 6. Image 1 (lower visual complexity).

were selected to differ in terms of visual complexity. The two stimuli were selected from the study by Tuch et al. (2009). In their research, visual complexity of an image was determined by the size of the JPEG file (in MB) and a pre-study validation survey. For the current study, two images of screenshots of academic websites were selected from images provided by Tuch et al. (2009).

The first stimulus image had a smaller file size (416 MB, Figure 6) than the second image (584 MB, Figure 7). Based on the pre-study survey data ($N = 57$, Wilcoxon signed-rank test), the first image was perceived as visually less complex (Image 1: $M = 2.4$, Image 2: $M = 4.3$, $Z = 5.36$, $p = <0.001$), more organized (Image 1: $M = 5.9$, Image 2: $M = 4.2$, $Z = 5.12$, $p = <0.001$) and less overloading (Image 1: $M = 1.4$, Image 2: $M = 4.0$, $Z = 6.43$, $p = <0.001$). Further, in the study by Tuch et al. (2009) the first stimulus was later remembered better in a one-week-delayed recognition task (mean recognition score 9.85) than the second stimulus image (5.79).

6.2. Participants, design, apparatus, and procedure

No specific selection criteria were applied in recruiting the 24 participants for the study. Thirteen female and eleven male participants with an average age of 42 (range 20–70 years) were selected based on convenience sampling. Most of the respondents had either a graduate degree (46%) or an undergraduate degree (33%). All the participants had a normal or corrected-to-normal vision, and they were compensated with a movie ticket. They were informed that they could withdraw from the study at any time.

The quasi-experimental study was conducted as a 2×4 within-subjects design. The participants responded to both stimulus with all four questionnaires under the control of an experimenter for counterbalancing possible order effects. The order of the images and the questionnaires was varied so that the order of both stimuli and questionnaires was counterbalanced over the total sample. The participants answered the questionnaires in a quiet, private space without external disturbances. The stimuli were presented to the participants on iPad with a screen size of 19.71 cm times 14.78 cm and a resolution of 1024×768 pixels. The four questionnaires were implemented as Webropol online surveys and answered with a laptop computer. A participant answered all the surveys in 30–45 min.

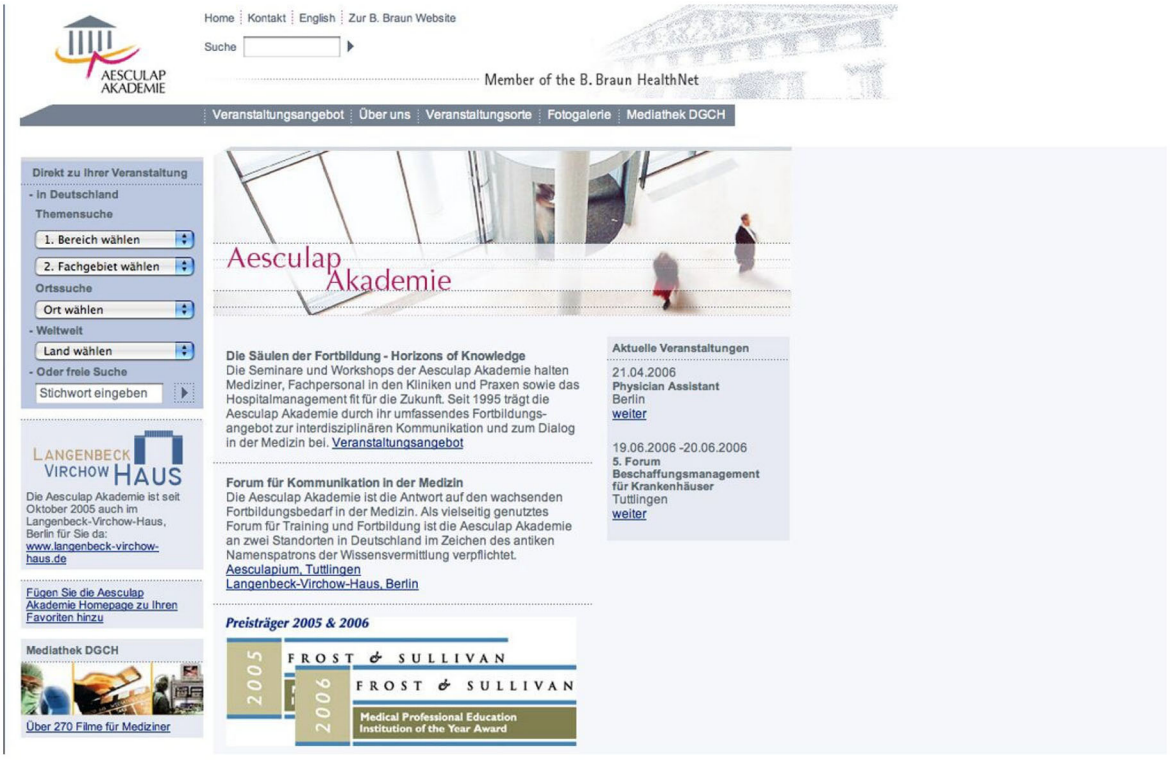


Figure 7. Image 2 (higher visual complexity).

Table 4. Cronbach's alphas for each questionnaire component per stimulus.

Questionnaire	Component	No. of items	α – Image 1	α – Image 2
VU	Visual clarity	5	0.71	0.87
	Composition	5	0.91	0.83
	Colors	5	0.63	0.74
VW	Simplicity	5	0.87	0.87
	Diversity	5	0.88	0.85
	Colorfulness	4	0.70	0.85
	Craftmanship	4	0.88	0.85
CEA	Classical aesthetics	5	0.89	0.85
	Expressive aesthetics	5	0.77	0.79
AD2	Pragmatic quality	7	0.80	0.93
	Hedonic quality (identity)	7	0.94	0.90
	Hedonic quality (stimulation)	7	0.90	0.93

VU: visual usability; VW: VisaWi; CEA: classical and expressive aesthetics; AD2: AttrakDiff2.

6.3. Analysis

The component scores were calculated as unweighted means of the contributing items. The reliability of the components was assessed by Cronbach's alpha. Due to mainly non-Gaussian distributions (Shapiro-Wilk $p < 0.05$) of the component scores, the measurements' ability to discriminate between the stimuli was assessed by nonparametric tests (Wilcoxon signed-rank test). Statistically significant differences together with effect sizes for the significant differences are reported to analyze and compare the sensitivity of the scale component for the differences in complexity in the stimuli. The statistical analyses were conducted with IBM SPSS (version 29).

6.4. Results

Cronbach's alphas for the components of each questionnaire are listed in Table 4. For all the questionnaires, the reliability of the components was close to acceptable or at good levels (all at or above 0.63).

Between the stimuli, there were significant differences observed in Visual clarity (VU, $Z = 3.55$, $p = 0.0001$, $d = 1.4$), Composition (VU, $Z = 2.86$, $p = 0.004$, $d = 0.8$), Simplicity (VW, $Z = 2.76$,

Table 5. Significant differences between the stimuli (Wilcoxon signed-rank test) and effect sizes as Cohen's *d*, $N = 24$.

Scale	Component	M (SD) – Image 1	M (SD) – Image 2	*Z*	*p*	*d*
VU	Visual clarity	4.03 (0.81)	2.68 (1.08)	3.55	.0001	1.4
	Composition	3.74 (1.16)	2.93 (0.95)	2.86	.004	0.8
VW	Simplicity	3.67 (1.01)	2.78 (1.00)	2.76	.006	0.9
CEA	Classical aesthetics	3.61 (0.98)	3.00 (0.94)	2.32	.021	0.6
AD2	Pragmatic quality	5.11 (0.86)	4.11 (1.43)	2.83	.005	0.9

$p = 0.0001$, $d = 0.9$), Classical aesthetics (CEA, $Z = 2.32$, $p = 0.021$, $d = 0.6$), and Pragmatic quality (AD2, $Z = 2.83$, $p = 0.005$, $d = 0.9$). All the significant differences were in favor of the visually less complex stimulus Image 1 (Figure 6, see Table 5). No significant differences were observed between the stimuli on Colors (VU, $p = 0.807$), Diversity (VW, $p = 0.999$), Colorfulness (VW, $p = 0.257$), Craftmanship (VW, $p = 0.346$), Expressive aesthetics (CEA, $p = 0.099$), Hedonic quality – identity (AD2, $p = 0.076$), or Hedonic quality – stimulation (AD2, $p = 0.145$).

7. General discussion

Exploratory factor analysis (EFA) provided support for the targeted psychometric structure of the VU scales. The comparative analysis between VU and visual experience measurements for 33 different websites ($N = 660$) indicated that the VU can reliably differentiate between visual usability and aesthetic experience in web design.

The strong correlation ($r = 0.55$) between the VU and AES – the latter measuring aesthetic appeal and expressive aesthetics – indicate strong relationships between visual usability and aesthetic appeal. This result strengthens previous research results concerning the relationship between perceived usability and aesthetic appeal (e.g., Sonderegger et al., 2012; Thüring & Mahlke, 2007; Tractinsky et al., 2006). This implies visual usability's ability to influence evaluations of aesthetic appeal, and possible moderating effects between perceived usability and aesthetic appeal, an important topic for future research. However, these relationships seem to vary based on the visual stimuli, as there were significant differences on the VU and AES ratings of various websites in the survey study (Figure 4).

The relationship between visual complexity and visual usability appears more straightforward than the relationship between these and aesthetic appeal. Both visual complexity and visual usability relate primarily to cognitive processing and information fluency than affective processes. Cognitively oriented evaluations are more unanimous than appraisals of aesthetic appeal (e.g., Silvennoinen & Jokinen, 2016a). Therefore, focusing on efficient visual usability increases the likelihood of positive, though less predictable aesthetic appeal, to occur. Thus, in measurements of visual technology experience, it is essential to be able to differentiate between the (more cognitively oriented sensemaking vehicle) components of visual usability and the (more affective and individually variable aesthetic appeal and appraisal) components of visual experience. This differentiation enables more predictability on the experience outcomes influenced by design decisions.

There was more variation in the AES scores than in the VU scores per stimuli, supporting previous research of people being more unanimous in visual usability evaluations than in assessments of aesthetic appeal (Silvennoinen & Jokinen, 2016a). Thus, focusing on designing efficient visual usability improves the probability of giving rise to positive aesthetic appraisals. The findings indicate that in evaluating visual usability of technological artifacts, the process seems to be more unified than in evaluations of aesthetic appeal. However, as these two dimensions influence each other, the more difficult dimension to design for (aesthetic appeal) can be enhanced by effective design decisions for visual usability. Visual usability as a sensemaking process can be considered as a cognitive dimension influencing aesthetic appeal (Silvennoinen & Jokinen, 2016a). In addition, designing for pleasurable and efficient visual usability is easier due to more established visual language of consisting components of visual representations compared to designing for other senses that lack detailed and established vocabularies as basis for verbalizing experiences pertaining to other senses than vision, and for scale development.

However, more research is needed to understand the inner workings of the aesthetic-usability effect, which can be approached by careful operationalization of the different dimensions of visual experience

and examinations of their interplay with validated scales. This is also the key to understanding possible moderating effects and what kinds of impact visual usability and aesthetic appeal have on user performance (e.g., Moshagen et al., 2009). The results of the current study indicate a need for further research on the moderating role of visual usability in the aesthetic-usability effect. In addition, according to the similarities and differences in the VU and aesthetic scores per website and the significant interaction between the measurement and the stimulus (Figure 4), it seems that the aesthetic-usability effect is highly context-dependent and dependent on the stimulus.

When manipulating visual complexity of two web pages, VU indicated the strongest effect size of the compared scales with the visual clarity measurement. Scales related to aesthetic appeal did not differentiate between the two stimuli. Other existing scales that showed significant differences were all related to visual usability (Table 5). The web pages did not differ from each other significantly in their color design, and the Colors factor did not show difference between them. The web page design was outdated (from 2009) which might have affected the aesthetic evaluations, but the two pages did not differ in these respects. At the item-level, there were no significant differences between the pages in the perceived up-to-datedness (VW, $p = 0.174$). The sample size was rather small ($N = 24$), but the observed significant and strong effects already with such a small sample indicate sensitivity of the novel scale for the manipulation of visual complexity. These findings suggest high construct validity of the new measurement and furthermore indicate that the new scale can provide more analytical understanding of the components of visual usability than the existing component scales focusing more on overall visual experience (simplicity, classical aesthetics, pragmatic quality). The overlap problem between the constructs is in their operationalizations, not in the inexistence of the overlap and interaction of these dimensions. For example, overall assessment of "professionalism" of websites requires both cognitive and affective evaluations.

The current study was limited to website designs, and the novel scale should be further developed and validated for different types of visual technological artifacts. Confirmatory factor analysis is the next logical step to further validate the scale, together with validating the survey in various languages. Only about third of the respondents were native English speakers and the measurement invariance in English and in translations must be still confirmed. In EFA, due to the low factor loading of the item focusing on contrast on the web page the item was removed from the color scale. As a plausible component of visual usability, more focused scale on perceived contrasts could therefore be developed to be included in the current VU scale. Based on the literature review, these components should ensure sufficient content validity of the novel scale. In future research, the visual usability scale could also be developed into an automated tool to provide researchers and practitioners with at least two ways of utilizing the measurement contents, as a survey and as an automated evaluation tool (cf., Michailidou et al., 2021).

8. Conclusions

The focus of this research was to bring clarity to the vastly studied phenomenon in HCI, visual experience and its underlying dimensions, such as aesthetic appeal. Even though research results indicate that aesthetic appeal influences perceived usability (e.g., Sonderegger et al., 2012; Thüring & Mahlke, 2007; Tractinsky, 2013; Tractinsky et al., 2000), various results of the dynamics have been reported. This stems from unsolid theoretical and conceptual grounds of the underlying dimensions, which lead to numerous operationalizations indicating differing results.

Therefore, a novel scale of a distinct sensemaking dimension of visual technology-experience, visual usability (VU), was created and validated. First, a systematic literature review was conducted to investigate how visual usability was defined in previous studies and through what kind of operationalizations it has been previously measured. Through this review, components pertaining to visual usability were detected and tested in two pilot studies and further elaborated for a validation study. Then, an exploratory factor analysis (EFA) was used to find and validate a psychometric structure for the VU factors, and analysis of variance (ANOVA) to study the differentiation of the VU scores from the scores of aesthetic appeal for 33 webpages ($N = 660$). Finally, the effects of visual complexity of websites on the VU and scales for measuring visual aesthetics were analyzed and compared.

Overall, the findings suggest that the novel scale can be reliably and effectively utilized for evaluating visual designs' visual usability and complexity regarding perceptions of visual clarity, unity, and color. Further, the findings indicate the need to differentiate visual usability measurement from measurements of aesthetic appeal to reliably examine the interplay of these underlying dimensions of visual experience.

The theoretical contribution of this study is in the conceptual clarification of the visual usability based on the literature review and the statistical analyses. Understanding what constitutes visual experience in HCI is broadened with the argumentation of the underlying dimensions of visual experience. Methodical contribution lies in the procedure of creating and validating a measurement for visual usability, which can be utilized in further examinations of the aesthetic-usability effect. Future research could utilize a similar procedure to create and validate a measurement applied in scrutinizing other essential concepts involved in visual experience in HCI, such as aesthetic appeal. In addition, visual usability could incorporate the existing W3C guidelines in terms of accessibility, as the VU measurement proposed here is a general one addressing issues concerning visual UI design.

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Ethical approval

The study was performed in accordance with the principles stated in the Declaration of Helsinki. Informed consent was obtained, and the participants were made aware that they can withdraw from the study at any time. All the data was anonymized. As the study adhered to the principles of the Finnish National Board on Research Integrity, ethical review by the Human Sciences Ethics Committee of the University of Jyväskylä was not required.

Author contributions

CRediT: **Johanna Silvennoinen**: Conceptualization, Data curation, Investigation, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing; **Tuomo Kujala**: Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Minna Haapakoski**: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Data will be made available upon request.

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